

MATH 2123 - MATHEMATICS III

2021 Exam Paper: Beginner Guide + Full Solutions

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Made for a beginner: learn the idea first, then follow the calculation.

Every set follows: Topic -> Basics -> Problem-solving procedure -> Full solution -> Exam tips.

How to use this guide

1. Read the Basics before looking at the numerical solution.
2. Memorize the Procedure as a reusable recipe for similar questions.
3. In the worked solution, understand why each formula is used instead of only copying it.
4. Before the exam, revise the green Final Answer boxes and the rapid-revision pages.

Source note

The questions were transcribed from the two photographs you provided. Handwritten marks on the paper were ignored; only the printed questions were solved.

Paper map - what the 8 sets are testing

Set	Main topic
1	Space curves: velocity, T-N-B frame, curvature, torsion, osculating plane and sketch
2	Vector-field classification, angle/tangent plane of surfaces, directional derivative, work line integral
3	Divergence theorem/flux, volume of a tetrahedral region, linear independence of vectors
4	Matrix definitions, simultaneous matrix equations, condition for inverse and inverse matrix
5	Laplace transform: periodic function, polynomial/exponential/trigonometric terms, $\sqrt{t}$ transforms, unit-step shift
6	Inverse Laplace: partial fractions, Heaviside expansion and logarithmic transforms
7	Convolution theorem, differential equation by Laplace, eigenvalues/eigenvectors
8	Non-square linear system and homogeneous system: row reduction and non-trivial solutions

The first four sets are Section A. Sets 5-8 are Section B. The strongest repeated themes are vector calculus, Laplace transforms, matrices and linear systems.

Symbols used in this guide

$\mathbf{r}(t)$ = position vector; $\mathbf{v}$ = velocity; T, N, B = unit tangent, normal and binormal.

$\text{div } \mathbf{F}$ = divergence; $\text{curl } \mathbf{F}$ = curl; $\text{grad } f$ = gradient.

$\mathcal{L}\{f(t)\}$ = Laplace transform and $\mathcal{L}^{-1}\{F(s)\}$ = inverse Laplace transform.

SET 1 - Space Curve Geometry

What this set tests

For $r(t) = 2 \cos t \mathbf{i} + 2 \sin t \mathbf{j} + k$: velocity, tangent, normal and binormal at $t=1$; radius of curvature and torsion at $t=0$; osculating plane at $t=1$; and a sketch of the curve.

Basics you need

- Velocity: $v = r'(t)$, speed = $|r'(t)|$.
- Unit tangent: $T = r'/|r'|$.
- Unit normal: $N = T'/|T'|$, provided T' is not zero.
- Binormal: $B = T \times N$.
- Curvature: $\kappa = |r' \times r''| / |r'|^3$. Radius of curvature $\rho = 1/\kappa$.
- Torsion: $\tau = [(r' \times r'') \cdot r'''] / |r' \times r''|^2$.
- Osculating plane is perpendicular to B : $B(t_0) \cdot [r - r(t_0)] = 0$.

Reusable procedure

Differentiate $r(t)$ three times. Then compute speed, T , N and B in that order.

For curvature use r' -prime cross r' -double-prime. For torsion take the scalar triple product with r' -triple-prime.

For the osculating plane, use B as the normal vector and the point $r(t_0)$.

1(a)(i) Velocity, T, N and B at $t=1$

Given

$$\begin{aligned}r(t) &= (2 \cos t, 2 \sin t, 1) \\r'(t) &= (-2 \sin t, 2 \cos t, 0) \\|r'(t)| &= \sqrt{4 \sin^2 t + 4 \cos^2 t} = 2 \\T(t) &= (-\sin t, \cos t, 0) \\T'(t) &= (-\cos t, -\sin t, 0), \quad |T'(t)| = 1 \\N(t) &= (-\cos t, -\sin t, 0) \\B(t) &= T \times N = (0, 0, 1)\end{aligned}$$

At $t=1$ radian:

$$\begin{aligned}v(1) &= (-2 \sin 1, 2 \cos 1, 0) \approx (-1.6829, 1.0806, 0) \\T(1) &= (-\sin 1, \cos 1, 0) \approx (-0.8415, 0.5403, 0) \\N(1) &= (-\cos 1, -\sin 1, 0) \approx (-0.5403, -0.8415, 0) \\B(1) &= (0, 0, 1)\end{aligned}$$

Final answer

$v(1) = (-2 \sin 1, 2 \cos 1, 0)$, $T(1) = (-\sin 1, \cos 1, 0)$, $N(1) = (-\cos 1, -\sin 1, 0)$, $B(1) = k$.

1(a)(ii) Radius of curvature and torsion at $t=0$

$$\begin{aligned}r''(t) &= (-2 \cos t, -2 \sin t, 0) \\r'''(t) &= (2 \sin t, -2 \cos t, 0) \\r' \times r'' &= (0, 0, 4), \text{ so } |r' \times r''| = 4 \\\kappa &= 4 / 2^3 = 1/2 \rightarrow \rho = 2 \\\tau &= [(r' \times r'') \cdot r'''] / |r' \times r''|^2 = 0/16 = 0\end{aligned}$$

Final answer

At $t=0$ (and in fact everywhere): radius of curvature $\rho=2$ and torsion $\tau=0$.

1(a)(iii) Osculating plane at $t=1$

The plane normal is $B(1)=(0,0,1)$. The point is $r(1)=(2 \cos 1, 2 \sin 1, 1)$.

$$(0,0,1) \cdot [(x,y,z) - (2 \cos 1, 2 \sin 1, 1)] = 0$$

$$z - 1 = 0$$

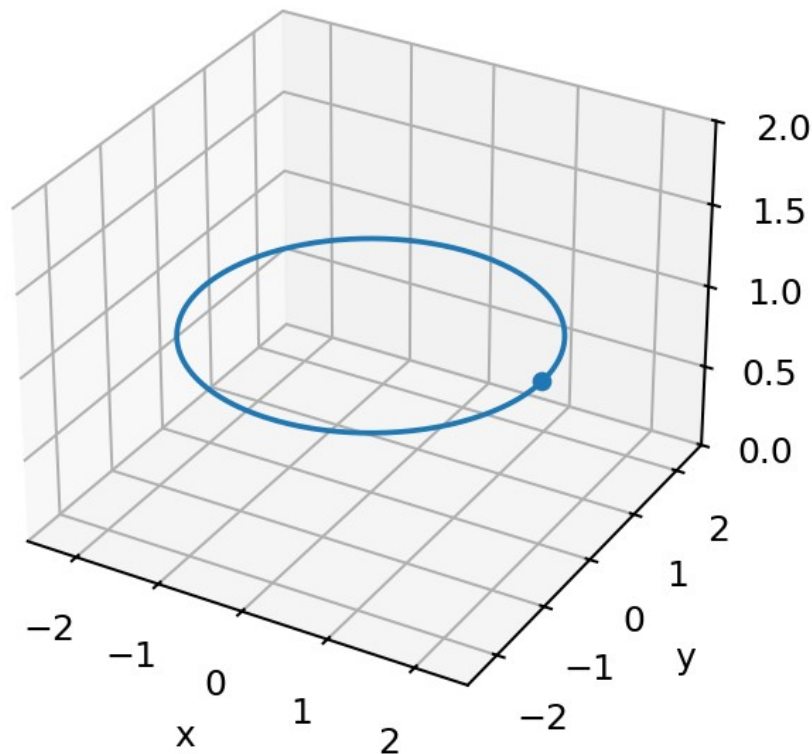
Final answer

Osculating plane: $z=1$.

1(a)(iv) What the curve looks like

Because $x=2 \cos t$, $y=2 \sin t$ and $z=1$, we have $x^2+y^2=4$ and $z=1$. So the particle moves around a circle of radius 2 in the horizontal plane $z=1$, centered at $(0,0,1)$. As t increases, it moves counterclockwise when viewed from $+z$.

Path: $x^2 + y^2 = 4$ in the plane $z = 1$



Common mistake / exam note

This is a planar circle, so torsion must be zero. That is an excellent quick check on the calculation.

SET 2 - Vector Fields, Surfaces and Work

What this set tests

- 2(a) Classify $F=r/r^2$ using div and curl.
- 2(b) Angle between two surfaces, tangent plane and directional derivative.
- 2(c) Work done by a force field along a parameterized curve.

Basics you need

- A vector field is solenoidal if $\text{div } F=0$.
- A vector field is irrotational if $\text{curl } F=0$. In a suitable simply connected region, it is conservative.
- For a surface $f(x,y,z)=\text{constant}$, $\text{grad } f$ is normal to the surface.
- Angle between surfaces = angle between their normal vectors (their gradients).
- Directional derivative in unit direction u : $D_u f = \text{grad } f \cdot u$.
- Work along $r(t)$: $W = \int F(r(t)) \cdot r'(t) dt$.

Reusable procedure

Classification: write components $\rightarrow$ compute div and curl $\rightarrow$ state solenoidal/irrotational.

Surfaces: compute gradients at the point $\rightarrow$ use dot product for the angle $\rightarrow$ use one gradient for tangent plane.

Work: substitute $x(t), y(t), z(t)$ into $F \rightarrow$ calculate $dr/dt \rightarrow$ dot $\rightarrow$ integrate over t .

2(a) Classify $F = r/r^2$

Let $R^2=x^2+y^2+z^2$. Then

$$F = (x/R^2, y/R^2, z/R^2), \quad R \neq 0$$

Divergence:

$$\text{div } F = d/dx(x/R^2) + d/dy(y/R^2) + d/dz(z/R^2) = 1/R^2$$

Therefore $\text{div } F$ is not zero (away from the origin), so it is not solenoidal.

For the curl, the cross partials cancel; equivalently

$$F = \text{grad}(\ln R)$$

$$\text{curl } F = \text{curl } \text{grad}(\ln R) = 0$$

Final answer

$F=r/r^2$ is irrotational (and conservative away from the origin), but it is not solenoidal. It is undefined at the origin.

Common mistake / exam note

Do not confuse r/r^2 with r/r^3 . The inverse-square radial field r/r^3 has zero divergence away from the origin; this paper has r/r^2 .

2(b) Angle of surfaces, tangent plane, directional derivative

Surfaces: $f_1=x^2+y^2+z^2=12$ and $f_2=xyz=8$ at $P=(2,2,2)$.

$$\text{grad } f_1=(2x, 2y, 2z) \rightarrow \text{grad } f_1(P)=(4, 4, 4)$$

$$\text{grad } f_2=(yz, xz, xy) \rightarrow \text{grad } f_2(P)=(4, 4, 4)$$

The normal vectors are parallel and in the same direction, so

$$\cos \theta = (n_1 \cdot n_2) / (|n_1| |n_2|) = 1 \rightarrow \theta=0$$

Final answer

Angle between the two surfaces at $(2,2,2)$: 0 degrees. They are tangent there.

Tangent plane to $x^2+y^2+z^2=12$ at $(2,2,2)$

$$4(x-2)+4(y-2)+4(z-2)=0$$

$$x+y+z=6$$

Final answer

Tangent plane: $x+y+z=6$.

Directional derivative of $f=xyz-8$ along the x-axis

$$\text{grad } f=(yz,xz,xy) \rightarrow \text{grad } f(2,2,2)=(4,4,4)$$

$$\text{Unit x-direction} = \mathbf{i}=(1,0,0)$$

$$D_{\mathbf{i}} f = \text{grad } f \cdot \mathbf{i} = 4$$

Final answer

Directional derivative along $+x$ at the point = 4.

2(c) Work along $x=t^2+1$, $y=2t^2$, $z=t^3$, $0 \leq t \leq 3$

Force field: $F=3xy \mathbf{i} - 5 \mathbf{j} + 10x \mathbf{k}$.

$$\mathbf{r}(t)=(t^2+1, 2t^2, t^3)$$

$$\mathbf{r}'(t)=(2t, 4t, 3t^2)$$

$$F(\mathbf{r}(t))=(6t^2(t^2+1), -5, 10(t^2+1))$$

$$F \cdot \mathbf{r}' = 12t^3(t^2+1) - 20t + 30t^2(t^2+1)$$

$$= 12t^5 + 30t^4 + 12t^3 + 30t^2 - 20t$$

$$W = \int_0^3 (\dots) dt$$

$$= [2t^6 + 6t^5 + 3t^4 + 10t^3 - 10t^2]_0^3$$

$$= 1458 + 1458 + 243 + 270 - 90 = 3339$$

Final answer

Work done = 3339 units.

SET 3 - Flux, Volume and Linear Independence

What this set tests

- 3(a) Flux through a closed quarter-cylinder using the divergence theorem.
- 3(b) Volume of a region cut from the first octant by a plane.
- 3(c) Linear dependence/independence of three vectors.

Basics you need

- Divergence theorem: closed-surface flux integral = triple integral of $\text{div } F$ over the enclosed volume.
- For a quarter-cylinder $x^2+y^2 \leq R^2$ in the first quadrant, use cylindrical coordinates: $x=r \cos \theta$, $y=r \sin \theta$, $0 \leq \theta \leq \pi/2$.
- A plane $x/a+y/b+z/c=1$ cuts a tetrahedron of volume $abc/6$ in the first octant.
- Three 3D vectors are linearly independent if the determinant formed from them is nonzero.

Reusable procedure

Closed flux: compute divergence $\rightarrow$ describe the volume $\rightarrow$ integrate in convenient coordinates.
First-octant plane: find x -, y -, z -intercepts $\rightarrow$ use tetrahedron volume formula.
Independence: put vectors as columns (or rows) $\rightarrow$ determinant. Nonzero means independent.

3(a) Closed-surface flux

The closed surface bounds the quarter-cylinder $x^2+y^2 \leq 9$, $x \geq 0$, $y \geq 0$, $0 \leq z \leq 2$.

Vector field: $q = (4yx^2, 2y^2z, -z^2)$.

$$\begin{aligned}\text{div } q &= d/dx(4yx^2) + d/dy(2y^2z) + d/dz(-z^2) \\ &= 8xy + 4yz - 2z\end{aligned}$$

Use $x=r \cos \theta$, $y=r \sin \theta$, $dV=r \, dr \, d\theta \, dz$ with $0 \leq r \leq 3$, $0 \leq \theta \leq \pi/2$, $0 \leq z \leq 2$.

$$\text{Flux} = \text{triple integral} [8xy + 4yz - 2z] \, dV$$

Evaluate term by term:

$$\text{Integral}(8xy) = 162$$

$$\text{Integral}(4yz) = 72$$

$$\text{Integral}(-2z) = -9\pi$$

$$\text{Total} = 162 + 72 - 9\pi = 234 - 9\pi$$

Final answer

Outward flux through the closed surface = $234 - 9\pi$.

Common mistake / exam note

Because S is explicitly closed, the divergence theorem is much easier than integrating separately over the cylinder wall, planes $x=0/y=0$, and top/bottom.

3(b) Volume bounded by $2x+3y+z=1$ and the coordinate planes

Find the intercepts:

$$x\text{-intercept: } y=z=0 \rightarrow x=1/2$$

$$y\text{-intercept: } x=z=0 \rightarrow y=1/3$$

$$z\text{-intercept: } x=y=0 \rightarrow z=1$$

$$\text{Volume} = abc/6 = (1/2)(1/3)(1)/6 = 1/36$$

Final answer

Volume = $1/36$ cubic unit.

3(c) Test the vectors for linear independence

$r_1=(3,1,-2)$, $r_2=(1,3,1)$, $r_3=(1,-5,0)$.

$$\det \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & -5 \\ -2 & 1 & 0 \end{bmatrix} = 32$$

Since 32 is not zero, the only solution of $c_1 r_1 + c_2 r_2 + c_3 r_3 = 0$ is $c_1 = c_2 = c_3 = 0$.

Final answer

The three vectors are linearly independent. Therefore there is no dependent vector to express as a combination of the others.

SET 4 - Matrix Types, Matrix Equations and Inverse

What this set tests

- 4(a) Definitions and examples: skew-symmetric, singular and diagonal matrices.
- 4(b) Solve two simultaneous matrix equations for A and B.
- 4(c) Decide whether a matrix is invertible and find its inverse.

Basics you need

- Skew-symmetric: $A^T = -A$. Every diagonal entry must be zero.
- Singular: $\det(A) = 0$. Such a square matrix has no inverse.
- Diagonal: all off-diagonal entries are zero.
- A square matrix is invertible iff $\det(A)$ is nonzero (equivalently $\text{rank} = n$).

Reusable procedure

Matrix definitions: state the defining condition and give a small example.

Simultaneous matrix equations: treat A and B like algebraic unknowns; eliminate one matrix.

Inverse: calculate determinant first. If nonzero, use Gauss-Jordan or $A^{-1} = \text{adj}(A)/\det(A)$.

4(a) Definitions with examples

(i) Skew-symmetric matrix

$$A^T = -A$$

Example: $\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$

(ii) Singular matrix

$$\det(A) = 0$$

Example: $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, because determinant = $4 - 4 = 0$

(iii) Diagonal matrix

Example: $\text{diag}(2, -1, 5) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

4(b) Find A and B

Given

$$3A - 2B = C = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}$$

$$-4A + B = D = \begin{bmatrix} -1 & 2 \\ -4 & 3 \end{bmatrix}$$

From the second equation:

$$B = D + 4A$$

Substitute into the first:

$$3A - 2(D + 4A) = C$$

$$-5A - 2D = C \rightarrow A = -(C + 2D)/5$$

$$C + 2D = \begin{bmatrix} 0 & 5 \\ -10 & 5 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & -1 \\ 2 & -1 \end{bmatrix}$$

$$B = D + 4A = \begin{bmatrix} -1 & -2 \\ 4 & -1 \end{bmatrix}$$

Final answer

$A = \begin{bmatrix} 0 & -1 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ 4 & -1 \end{bmatrix}$.

4(c) Inverse of the matrix

$$M = \begin{bmatrix} 2 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$$

$$\det(M) = -10 \neq 0$$

So the inverse exists. The adjugate is

$$\text{adj}(M) = \begin{bmatrix} -11 & 2 & 2 \\ -4 & 8 & -2 \\ 6 & -2 & -2 \end{bmatrix}$$

$$M^{-1} = \text{adj}(M) / (-10)$$

$$= \begin{bmatrix} 11/10 & -1/5 & -1/5 \\ 2/5 & -4/5 & 1/5 \\ -3/5 & 1/5 & 1/5 \end{bmatrix}$$

Final answer

The matrix is nonsingular and its inverse is $\begin{bmatrix} 11/10 & -1/5 & -1/5 \\ 2/5 & -4/5 & 1/5 \\ -3/5 & 1/5 & 1/5 \end{bmatrix}$.

Common mistake / exam note

Always test $\det(A)$ before starting a long inverse calculation.

SET 5 - Laplace Transform Techniques

What this set tests

- 5(a) Definition of Laplace transform, periodic functions, and transforms of t^n , e^{at} and $t^2 \cos(2t)$.
- 5(b) Nonstandard transforms involving $\sin(\sqrt{t})$ and $\cos(\sqrt{t})/\sqrt{t}$.
- 5(c) Unit-step/time-shift form.

Basics you need

- Definition: $L\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$.
- If f has period T : $L\{f\} = [\int_0^T e^{-st} f(t) dt] / [1 - e^{-sT}]$.
- $L\{t^n\} = n! / s^{n+1}$, $L\{e^{-at}\} = 1 / (s+a)$.
- $L\{t^n f(t)\} = (-1)^n d^n F / ds^n$.
- Second shifting: $L\{u(t-a) g(t-a)\} = e^{-as} G(s)$.

Reusable procedure

Periodic: integrate only one period and divide by $1 - e^{-sT}$.

Mixed expression: split by linearity, use table entries, and use differentiation in s for factors of t .

Delayed function: rewrite exactly as $u(t-a)g(t-a)$ and apply e^{-as} .

5(a)(i) Periodic function

$F(t) = 3t$ for $0 < t < 2$, $F(t) = 6$ for $2 < t < 4$, with period $T = 4$.

$$L\{F\} = [\text{Integral}_0^2 3t e^{-st} dt + \text{Integral}_2^4 6e^{-st} dt] / (1 - e^{-4s})$$

$$\text{Integral}_0^2 3t e^{-st} dt = 3[1 - (1+2s)e^{-2s}] / s^2$$

$$\text{Integral}_2^4 6e^{-st} dt = 6[e^{-2s} - e^{-4s}] / s$$

The e^{-2s}/s terms cancel, giving

$$L\{F\} = [3(1 - e^{-2s}) / s^2 - 6e^{-4s} / s] / [1 - e^{-4s}]$$

Final answer

$$L\{F(t)\} = \{3(1 - e^{-2s}) - 6s e^{-4s}\} / \{s^2(1 - e^{-4s})\}.$$

5(a)(ii) $F(t) = 3t^4 + 4e^{-3t} + t^2 \cos 2t$

$$L\{3t^4\} = 3(4!) / s^5 = 72 / s^5$$

$$L\{4e^{-3t}\} = 4 / (s+3)$$

For the last term, start from $L\{\cos 2t\} = s / (s^2 + 4)$. Since t^2 gives two s -derivatives:

$$L\{t^2 \cos 2t\} = d^2 / ds^2 [s / (s^2 + 4)]$$

$$= 2s(s^2 - 12) / (s^2 + 4)^3$$

Final answer

$$L\{F(t)\} = 72/s^5 + 4/(s+3) + 2s(s^2 - 12)/(s^2 + 4)^3.$$

5(b) $L\{\sin \sqrt{t}\}$ and $L\{\cos \sqrt{t}/\sqrt{t}\}$

These look unusual. Use $t = u^2$, so $dt = 2u du$. A standard Gaussian integral is

$$\text{Integral}_0^\infty e^{-s u^2} \cos(bu) du = \sqrt{\pi} / (2\sqrt{s}) * e^{-b^2/(4s)}$$

Differentiate this with respect to b to get the $u \sin(bu)$ integral.

(i) $\sin \sqrt{t}$

$$\begin{aligned} L\{\sin \sqrt{t}\} &= \int_0^{\infty} e^{-st} \sin(\sqrt{t}) dt \\ &= 2 \int_0^{\infty} u e^{-s u^2} \sin u du \\ &= \sqrt{\pi}/(2 s^{3/2}) e^{-1/(4s)} \end{aligned}$$

(ii) $\cos \sqrt{t}/\sqrt{t}$

$$\begin{aligned} L\{\cos \sqrt{t}/\sqrt{t}\} &= 2 \int_0^{\infty} e^{-s u^2} \cos u du \\ &= \sqrt{\pi/s} e^{-1/(4s)} \end{aligned}$$

Final answer

$$L\{\sin \sqrt{t}\} = \sqrt{\pi}/(2s^{3/2}) e^{-1/(4s)}; \quad L\{\cos \sqrt{t}/\sqrt{t}\} = \sqrt{\pi/s} e^{-1/(4s)}.$$

5(c) Delayed sine

$F(t)=5 \sin[3(t-\pi/4)]$ for $t>\pi/4$ and $F(t)=0$ for $t<\pi/4$.

$$F(t)=u(t-\pi/4) * 5 \sin[3(t-\pi/4)]$$

$$L\{5 \sin 3t\}=15/(s^2+9)$$

$$L\{F\}=e^{(-\pi s/4)} * 15/(s^2+9)$$

Final answer

$$L\{F(t)\} = 15 e^{(-\pi s/4)}/(s^2+9).$$

SET 6 - Inverse Laplace Transform

What this set tests

- 6(a) Partial fractions with a linear and an irreducible quadratic factor.
- 6(b) Inverse transform using Heaviside/cover-up style expansion.
- 6(c) Standard inverse transforms plus a logarithmic transform.

Basics you need

- Factor the denominator first. For an irreducible quadratic, use $(Bs+C)/(s^2+a^2)$.
- $L^{-1}\{s/(s^2+a^2)\}=\cos(at)$, $L^{-1}\{a/(s^2+a^2)\}=\sin(at)$.
- $L^{-1}\{1/(s-a)\}=e^{at}$.
- Useful log identity: $L\{(1-e^{-at})/t\}=\ln((s+a)/s)$.

Reusable procedure

Rational function: partial fraction $\rightarrow$ standard table $\rightarrow$ inverse term by term.

Heaviside: use cover-up immediately for simple linear factors; then find the remaining numerator coefficients.

Logarithm: factor the expression inside $\ln$ and use a known logarithmic transform identity.

6(a) Invert $(s+5)/[(s+1)(s^2+4)]$

$$(s+5)/[(s+1)(s^2+4)] = A/(s+1) + (Bs+C)/(s^2+4)$$

$$A=4/5, B=-4/5, C=9/5$$

$$F(s)=4/[5(s+1)] - (4/5)s/(s^2+4) + (9/5)/(s^2+4)$$

$$L^{-1}\{1/(s^2+4)\}=(1/2)\sin 2t$$

Final answer

$$f(t)=(4/5)e^{-t} - (4/5)\cos 2t + (9/10)\sin 2t.$$

6(b) Invert $(3s+1)/[(s-1)(s^2+1)]$

Use cover-up on the factor $(s-1)$:

$$A = [(3s+1)/(s^2+1)] \text{ at } s=1 = 4/2 = 2$$

Write the rest as $(Bs+C)/(s^2+1)$. Matching coefficients gives $B=-2$, $C=1$.

$$F(s)=2/(s-1) - 2s/(s^2+1) + 1/(s^2+1)$$

Final answer

$$f(t)=2e^t - 2\cos t + \sin t.$$

6(c)(i) Invert $(s^2+3)/[s(s^2+9)]$

$$(s^2+3)/[s(s^2+9)] = A/s + Bs/(s^2+9)$$

$$9A=3 \rightarrow A=1/3; \quad A+B=1 \rightarrow B=2/3$$

Final answer

$$f(t)=1/3 + (2/3)\cos 3t.$$

6(c)(ii) Invert $\log[(s^2-1)/s^2]$

Factor the logarithm:

$$\ln[(s^2-1)/s^2] = \ln[(s+1)/s] + \ln[(s-1)/s]$$

Use $L\{(1-e^{-at})/t\} = \ln[(s+a)/s]$.

$$\ln[(s+1)/s] \leftrightarrow (1-e^{-t})/t$$

$$\ln[(s-1)/s] \leftrightarrow (1-e^t)/t$$

$$\text{Sum} = [2-e^{-t}-e^t]/t = 2(1-\cosh t)/t$$

Final answer

$L^{-1}\{\ln[(s^2-1)/s^2]\} = 2(1-\cosh t)/t$, with the transform understood for $\text{Re}(s) > 1$.

SET 7 - Convolution, ODE by Laplace and Eigenvalues

What this set tests

- 7(a) Convolution theorem for a product of two cosine transforms.
- 7(b) Solve a second-order initial-value problem using Laplace transform.
- 7(c) Eigenvalues and eigenvectors of a 2x2 matrix.

Basics you need

- Convolution: if $F=L\{f\}$, $G=L\{g\}$, then $L^{-1}\{FG\}=\int_0^t f(u)g(t-u)du$.
- $L\{y'\}=sY-y(0)$; $L\{y''\}=s^2Y-sy'(0)-y''(0)$.
- Eigenvalues solve $\det(A-\lambda I)=0$. For each eigenvalue, solve $(A-\lambda I)v=0$.

Reusable procedure

Convolution: split the product into recognizable transforms $\rightarrow$ write integral $\rightarrow$ simplify.

Laplace ODE: transform every term $\rightarrow$ insert initial conditions immediately $\rightarrow$ solve algebraically for $Y(s)$ $\rightarrow$ partial fractions $\rightarrow$ inverse.

Eigen problem: characteristic determinant $\rightarrow$ roots $\rightarrow$ nullspace for each root.

7(a) Using convolution theorem

Find $L^{-1}\{s^2/[(s^2+1)(s^2+2)]\}$.

$$s^2/[(s^2+1)(s^2+2)] = [s/(s^2+1)] [s/(s^2+2)]$$

$$L^{-1}\{s/(s^2+1)\}=\cos t$$

$$L^{-1}\{s/(s^2+2)\}=\cos(\sqrt{2}t)$$

$$f(t)=\int_0^t \cos u * \cos[\sqrt{2}(t-u)] du$$

Evaluating the convolution integral gives

$$f(t)=\sqrt{2} \sin(\sqrt{2}t) - \sin t$$

Final answer

$$L^{-1}\{s^2/[(s^2+1)(s^2+2)]\} = \sqrt{2}\sin(\sqrt{2}t) - \sin t.$$

Common mistake / exam note

A quick check by partial fractions gives $-1/(s^2+1)+2/(s^2+2)$, which inverts to the same result. In the exam, still show the convolution integral because the question explicitly asks for convolution theorem.

7(b) Solve $y''-3y'+2y=e^{2t}$, $y(0)=-3$, $y'(0)=5$

Let $Y(s)=L\{y(t)\}$.

$$L\{y''\}=s^2Y-s(-3)-5 = s^2Y+3s-5$$

$$L\{-3y'\}=-3[sY-(-3)] = -3sY-9$$

$$L\{2y\}=2Y, \quad L\{e^{2t}\}=1/(s-2)$$

$$(s^2-3s+2)Y + 3s-14 = 1/(s-2)$$

$$Y = -10/(s-1) + 7/(s-2) + 1/(s-2)^2$$

$$L^{-1}\{1/(s-a)^2\}=t e^{at}$$

Final answer

$$y(t)=-10e^t + 7e^{2t} + t e^{2t}.$$

7(c) Eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

$$\begin{aligned}\det(A - \lambda I) &= (1 - \lambda)(3 - \lambda) - 8 \\ &= \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1)\end{aligned}$$

Thus $\lambda_1 = 5$, $\lambda_2 = -1$.

For $\lambda = 5$

$$A - 5I = \begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix} \rightarrow -4x + 4y = 0 \rightarrow x = y$$

One eigenvector: $v_1 = (1, 1)^T$

For $\lambda = -1$

$$A + I = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix} \rightarrow 2x + 4y = 0 \rightarrow x = -2y$$

One eigenvector: $v_2 = (-2, 1)^T$

Final answer

Eigenpairs: $\lambda = 5$ with $v = (1, 1)^T$; $\lambda = -1$ with $v = (-2, 1)^T$. Any nonzero scalar multiple is also valid.

SET 8 - Linear Systems and Non-trivial Solutions

What this set tests

- 8(a) Solve a 3-equation, 4-unknown system by row reduction; expect a free variable.
8(b) Find all non-trivial solutions of a homogeneous 3x3 system.

Basics you need

- Row operations: swap rows; multiply a row by a nonzero constant; add a multiple of one row to another.
- A pivot variable is determined by the equations; a non-pivot variable is free.
- A homogeneous system $Ax=0$ always has the trivial solution. Non-trivial solutions exist if $\text{rank}(A) < \text{number of unknowns}$ (equivalently $\det(A)=0$ for a square matrix).

Reusable procedure

Form the augmented matrix $\rightarrow$ row-reduce to RREF $\rightarrow$ identify pivot and free variables $\rightarrow$ write parametric solution.

For homogeneous systems, row-reduce the coefficient matrix. Give the whole solution family, not just one vector.

8(a) Solve the system

$$x_1 - 2x_2 + 3x_3 - 4x_4 = 5$$

$$x_1 - 3x_2 + 7x_3 - 6x_4 = 7$$

$$2x_1 - 5x_2 + 10x_3 - 11x_4 = 12$$

The augmented matrix row-reduces to

$$\text{RREF} = \left[\begin{array}{cccc|c} 1 & 0 & -5 & 0 & 1 \\ 0 & 1 & -4 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

So

$$x_1 - 5x_3 = 1, \quad x_2 - 4x_3 = -2, \quad x_4 = 0$$

Let the free variable $x_3 = a$.

$$x_1 = 1 + 5a, \quad x_2 = -2 + 4a, \quad x_3 = a, \quad x_4 = 0$$

Final answer

All solutions: $(x_1, x_2, x_3, x_4) = (1 + 5a, -2 + 4a, a, 0)$, where a is any real number.

8(b) Find all non-trivial solutions

$$3x_1 + 5x_2 - 2x_3 = 0$$

$$x_1 - 3x_2 + 4x_3 = 0$$

$$2x_1 + x_2 + x_3 = 0$$

Row-reducing the coefficient matrix gives

$$\text{RREF} = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_1 + x_3 = 0, \quad x_2 - x_3 = 0$$

Let $x_3 = a$. Then $x_1 = -a$ and $x_2 = a$.

$$(x_1, x_2, x_3) = a(-1, 1, 1)$$

Final answer

All solutions are $a(-1, 1, 1)$. Non-trivial solutions correspond to any $a \neq 0$.

Rapid Revision - Reusable Procedures

Space curve (T-N-B)

1. $r' \rightarrow \text{speed} \rightarrow T$.
2. $T' \rightarrow N$.
3. $B = T \times N$.
4. $\kappa = |r' \times r''|/|r'|^3$; $\rho = 1/\kappa$.
5. $\tau = [(r' \times r'') \cdot r''']/|r' \times r''|^2$.
6. Osculating plane uses B as normal.

Vector calculus

Classification: div for solenoidal, curl for irrotational.

Surface normal = gradient of its defining scalar function.

Work = $\int F \cdot dr$. Closed-surface flux $\rightarrow$ try divergence theorem first.

Matrices and systems

$\det(A) \neq 0 \rightarrow$ inverse exists and square system has a unique solution.

Linear independence in R^3 : determinant of three vectors $\neq 0$.

Row-reduce systems. Non-pivot variables are free parameters.

Laplace

Periodic: one-period integral divided by $1 - e^{-sT}$.

Multiplying by $t^n \rightarrow$ differentiate $F(s)$ n times with alternating sign.

Delay $u(t-a)g(t-a) \rightarrow e^{-as}G(s)$.

Inverse rational function $\rightarrow$ partial fractions. Product $\rightarrow$ convolution when requested.

ODE $\rightarrow$ transform derivatives, insert initial values, solve for $Y(s)$, invert.

Final Answers at a Glance

Question	Answer
1	$T = (-\sin t, \cos t, 0)$, $N = (-\cos t, -\sin t, 0)$, $B = k$; at $t=1$ substitute $t=1$. $\rho=2$, $\tau=0$. Osculating plane $z=1$. Curve: circle $x^2+y^2=4$ in $z=1$.
2(a)	$F=r/r^2$ is irrotational/conservative away from origin, not solenoidal; $\text{div } F=1/r^2$, $\text{curl } F=0$.
2(b)	Angle=0; tangent plane $x+y+z=6$; directional derivative of $xyz-8$ along $+x = 4$.
2(c)	Work=3339.
3(a)	Closed flux=234-9pi.
3(b)	Volume=1/36.
3(c)	Vectors are independent; determinant=32.
4(b)	$A = \begin{bmatrix} 0 & -1 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ 4 & -1 \end{bmatrix}$.
4(c)	$A^{-1} = \begin{bmatrix} 11/10 & -1/5 & -1/5 \\ 2/5 & -4/5 & 1/5 \\ -3/5 & 1/5 & 1/5 \end{bmatrix}$.
5(a)(i)	$\{3(1-e^{-2s})-6s e^{-4s}\}/\{s^2(1-e^{-4s})\}$.

5(a)(ii)	$72/s^5 + 4/(s+3) + 2s(s^2-12)/(s^2+4)^3.$
5(b)	$\sqrt{\pi}/(2s^{3/2})e^{(-1/4s)}; \text{ and } \sqrt{\pi/s}e^{(-1/4s)}.$
5(c)	$15e^{(-\pi s/4)}/(s^2+9).$
6(a)	$(4/5)e^{-t} - (4/5)\cos 2t + (9/10)\sin 2t.$
6(b)	$2e^t - 2\cos t + \sin t.$
6(c)	$1/3 + (2/3)\cos 3t; \text{ and } 2(1 - \cosh t)/t.$
7(a)	$\sqrt{2}\sin(\sqrt{2}t) - \sin t.$
7(b)	$y = -10e^t + 7e^{(2t)} + t e^{(2t)}.$
7(c)	$\lambda = 5, v = (1, 1); \lambda = -1, v = (-2, 1).$
8(a)	$(1+5a, -2+4a, a, 0).$
8(b)	$a(-1, 1, 1), a \neq 0 \text{ for non-trivial solutions.}$

Exam Strategy for This Paper

If you are short on preparation time

1. Master Sets 4 and 8 first: matrix definitions, inverse and row reduction are procedural and score-friendly.
2. Learn the standard Laplace table and partial fractions, then practice Sets 5-7.
3. For vector calculus, memorize the meaning of grad/div/curl and the formulas for work and flux.
4. For space curves, memorize the order $T \rightarrow N \rightarrow B$ and the curvature/torsion formulas.
5. In the exam, write formulas before substituting values. Even if arithmetic goes wrong, correct method can earn partial marks.

One-page memory list

$T = \mathbf{r}'/|\mathbf{r}'|$; $N = \mathbf{T}'/|\mathbf{T}'|$; $B = T \times N$.

$\kappa = |\mathbf{r}' \times \mathbf{r}''|/|\mathbf{r}'|^3$; $\tau = \text{triple product}/|\text{cross product}|^2$.

$D_u f = \text{grad } f \cdot u$; $\text{work} = \int F \cdot dr$; $\text{flux}(\text{closed}) = \text{triple integral div } F \, dV$.

Inverse exists iff $\det \neq 0$. Eigenvalues: $\det(A - \lambda I) = 0$.

$L\{t^n\} = n!/s^{n+1}$; $L\{e^{at}\} = 1/(s-a)$; $L\{\sin at\} = a/(s^2+a^2)$; $L\{\cos at\} = s/(s^2+a^2)$.

End of 2021 beginner guide.